DATA ITEM DESCRIPTION
Title: MATERIALS AND PROCESSES SUBSTANTIATING DATA AND ANALYSIS TEST REPORT

Number: DI-SESS-82237A
AMSC Number: N10402
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20230510
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2023-021

Use/relationship: The Materials and Processes Substantiating Data and Analysis Test Report documents the data required to substantiate the materials and processes used to develop materiel. The data will be used to substantiate the materiel's design to ensure system performance requirements are met throughout the system's operational service life.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described by the contract SOW.

This DID interfaces with DI-SESS-82236A, Materials and Processes Substantiating Data and Analysis Test Plan.

This DID supersedes DI-SESS-82237.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. Contractor format acceptable.

3. Content. The Materials and Processes Substantiating Data and Analysis Test Report shall contain the following:

3.1. Testing Methods for Material Properties:

3.1.1. Describe the results from all research and development, testing, and fabrication studies accomplished with respect to materials and processes development.

3.1.2. Describe the results from conducting materials testing.

3.1.3. Describe the work performed to support the evaluation and substantiation of material properties.

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3.1.4. Describe the results from developing non-destructive test accept/reject criteria, materials processing effects and parameters, material fabrication techniques, and manufacturing procedures.

3.1.5. Describe the ability to meet program specific test objectives, test methods, test procedures, test conditions, and test matrices proposed in the Materials and Processes Substantiating Data and Analysis Test Plan.

3.1.6. Describe the data generated to validate the methods being used in the testing and evaluation of materials.

3.1.7. Describe the data generated from materials testing and evaluation to ensure that materials and processes are characterized in a statistically significant manner relative to the design application.

3.2. Material Properties Testing Plans and Procedures:

3.2.1. Describe the data available to substantiate that the testing and evaluation strategy supports the building block approach (coupon, element, sub component). Describe the rationale for using approaches other than the building block approach.

3.2.2. Describe the data available to substantiate the implementation of procedures used for monitoring and controlling processes critical to satisfying part design requirements.

3.2.3. Describe the process for identifying, monitoring, controlling, and analyzing key process parameters.

3.2.4. Describe the process for determining and evaluating process capability, including management and reduction of process variation.

3.2.5. Describe the methodology used to determine and substantiate thermal exposure time histories.

3.2.6. Describe the data to substantiate composite sandwich construction properties, including those analytically derived from sandwich constituent properties to permit for the statistical determination of the material properties.

3.2.7. Describe the methodology and results for process verification testing of test coupons (representative of the production part) and their associated acceptance criteria.

3.3. Material Properties Test Schedule: Describe the materials and processes testing schedule that includes the characterization, testing, and analysis planned with respect to critical path milestones against the contractor’s integrated master schedule.

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3.4. Material Properties Substantiation:

3.4.1. Describe the methods used to substantiate material properties by using standardized documentation, analyses, tests, and inspections to characterize the following properties: static strength, fatigue strength, modulus, effect of stress concentrations, fracture toughness, crack growth, impact resistance, effects of defects, resistance to environmentally-assisted cracking, wear, general corrosion, pitting, stress corrosion cracking, exfoliation, galvanic corrosion, crevice corrosion, filiform corrosion, moisture absorption, chemical, solvent, fuel, and lubricant exposure; hydrolytic instability, thermal exposure, electromagnetic compatibility, adhesion strength, peel resistance, dielectric constant, resistivity, glass transition temperature, and erosion.

3.4.2. Describe the data to substantiate the effects of environmental exposure and operating environment on material properties.

3.4.3. Describe the data to substantiate materials specifications selected for protective finishes, protective coatings and sealants for corrosion protection.

3.5. Material Allowables Development:

3.5.1. Describe the design allowable philosophy used to develop the system materials, and document how the data was generated and used to determine material properties.

3.5.2. Describe the analytical procedures used to calculate static and dynamic material allowables.

3.5.3. Describe the static material allowables methodology in statistical terms of confidence levels and exceedance values.

3.5.4. Describe the methodology used to substantiate the impact of inorganic finish systems on material allowables.

3.5.5. Describe the methods used to generate composite laminate data (including data analytically derived from single-ply properties) to permit for a statistical determination of the laminate properties.

3.5.6. Describe the data to substantiate the materials' fracture toughness values to be used in the residual strength analysis of metallic parts will be the minimum substantiated fracture toughness values.

End of DI-SESS-82237A

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